KIT - Kalaignarkarunanidhi Institute of Technology

(An Autonomous Institute)

Department of Computer Applications

M23CAP401– PROJECT WORK

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PROJECT TITLE:

Smart Answering System using Retrieval-Augmented Generation (RAG)

ABSTRACT

- Developed a Smart Answering System using Retrieval-Augmented Generation (RAG).
- Understands user questions in natural language.
- Retrieves relevant information from uploaded documents before generating answers.
- Converts documents into vectors for efficient semantic search.
- Stores vectors in a vector database for fast information retrieval.
- Uses an open-source Large Language Model (LLM) to generate context-based responses.
- Reduces hallucination by grounding answers in retrieved documents.
- Handles large volumes of data efficiently.
- Suitable for Researchers, Developers, Data Analysts, and Corporate Professionals.

EXISTING SYSTEM

- Uses only keyword-based search without semantic understanding.
- Primarily works with plain text and struggles to process PDFs or images effectively.
- Stores data in traditional databases without contextual understanding.
- Generates answers without validating information, leading to possible incorrect responses.
- Cannot efficiently handle large unstructured data such as scanned documents.
- Performance decreases as data size increases.
- Requires manual rule-based methods for information extraction.
- Provides less accurate answers for complex or multi-part queries.
- Not suitable for real-time intelligent applications.
- Does not utilize modern AI techniques or multimodal models.

PROPOSED SYSTEM

- Can understand text, images, PDFs, and scanned documents in one system
- Supports different file formats and adjusts automatically
- Uses open-source AI models with a smart agent to understand user questions
- Converts all data into vector form so the system can understand meaning
- Stores the data in a vector database for quick access
- Smart agent analyzes and improves the user question if needed
- Finds the most relevant information based on the question
- Thinks step-by-step to generate accurate answers
- Gives clear, explainable, and context-based responses
- Useful for chatbots, research, document analysis, and decision making

SYSTEM SPECIFICATION

HARDWARE

- Processor: Intel Core i5 or higher
- RAM: Minimum:16 GB
- Storage: Minimum 512 GB HDD/SSD
- System Type: 64-bit computer

SOFTWARE

- Operating System: Windows 11, Linux, or macOS
- Programming Language: Python 3.10 or above
- Database:MYSQL, Chroma Vector Database
- Tools: VS Code

SYSTEM MODULES

- Modules 1-Login page Module
- Modules 2-User Interface Module
- Modules 3-Query Input & Preprocessing Module
- Modules 4-Embedding Generation Module
- Modules 5-Information Retrieval Module
- Modules 6-Responses Generation Module
- Modules 7-Database Module

Module description

Modules 1-Login Page Module

- Authenticates users to securely access the Smart Answering System
- Stores login details in MySQL and enables controlled system usage

Modules 2-User Interface Module

- Provides interface to enter queries and view AI-generated answers
- Displays previous queries and responses for easy reference.

Modules 3 -Query Input & Preprocessing Module

- Processes user queries by cleaning and structuring input text
- Prepares queries for embedding to improve RAG-based retrieval accuracy

Modules 4-Embedding Generation Module

- Analyzes user intent and rewrites complex or unclear queries if necessary.
- Plans intelligent retrieval steps to improve answer accuracy.

Moduels 5-Information Retrieval Module

- Combines retrieved information with the user query to generate structured and context based responses.
- Ensures answers are meaningful, accurate, and grounded in the uploaded documents.

Modules 6-Response Generation Module

- Generates accurate answers using LLM based on retrieved context
- Combines query + documents to produce context-aware responses (RAG)

Modules 7-Database Module

- Stores user data, documents, embeddings, and responses
- Supports fast retrieval and efficient management of RAG pipeline data.

MODELS

Llama 3.1 8B (Large Language Model)

- Used for text understanding, reasoning, and response generation
- Powers RAG-based chat and intelligent answer synthesis

Qwen 2.5 VL (Vision-Language Model)

- Processes and analyzes images along with text
- Provides visual context understanding for multimodal queries

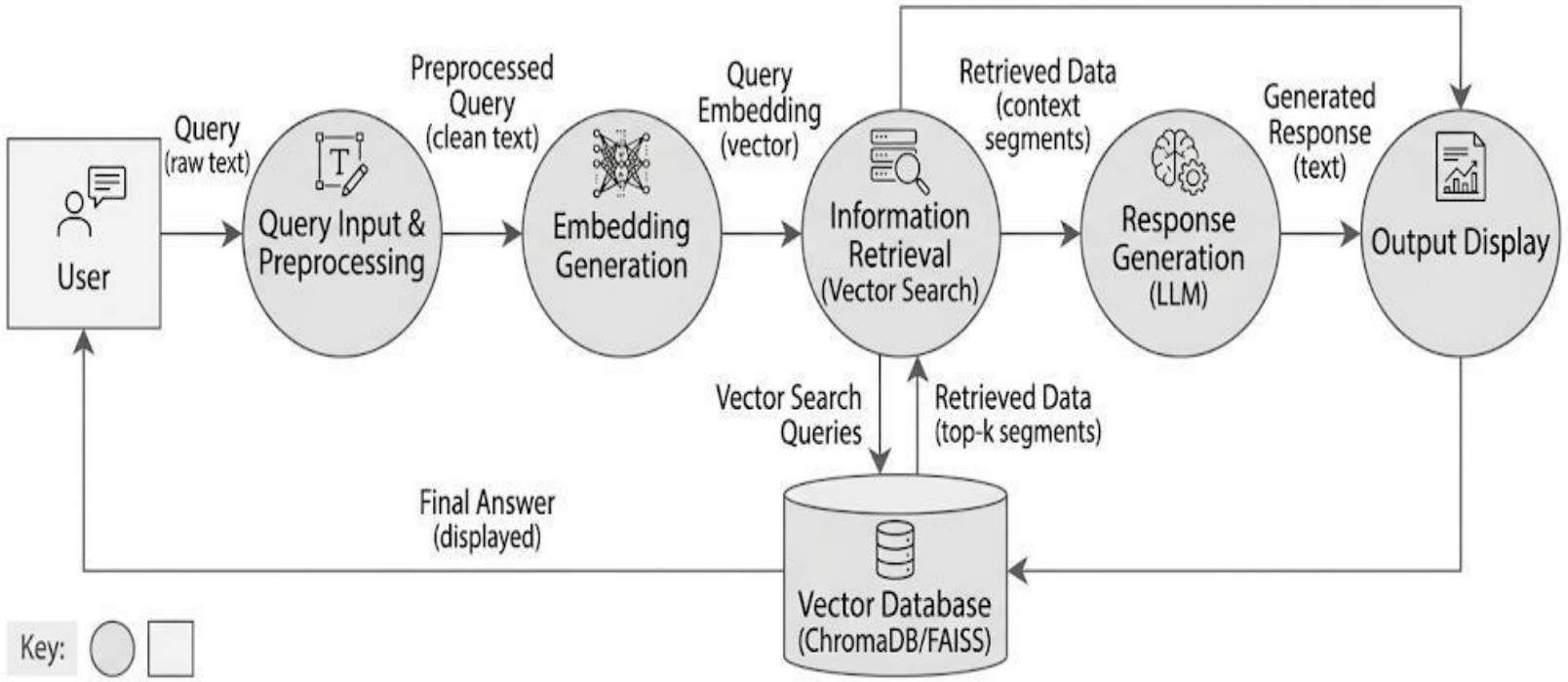
all-MiniLM-L6-v2 (Embedding Model)

- Converts text into vector representations
- Enables fast and accurate semantic search in the database

CLIP ViT-B/32 (Multimodal Embedding Model)

- Encodes images and text into a shared vector space
- Supports image search and cross-modal retrieval

DATA FLOW DIAGRAM



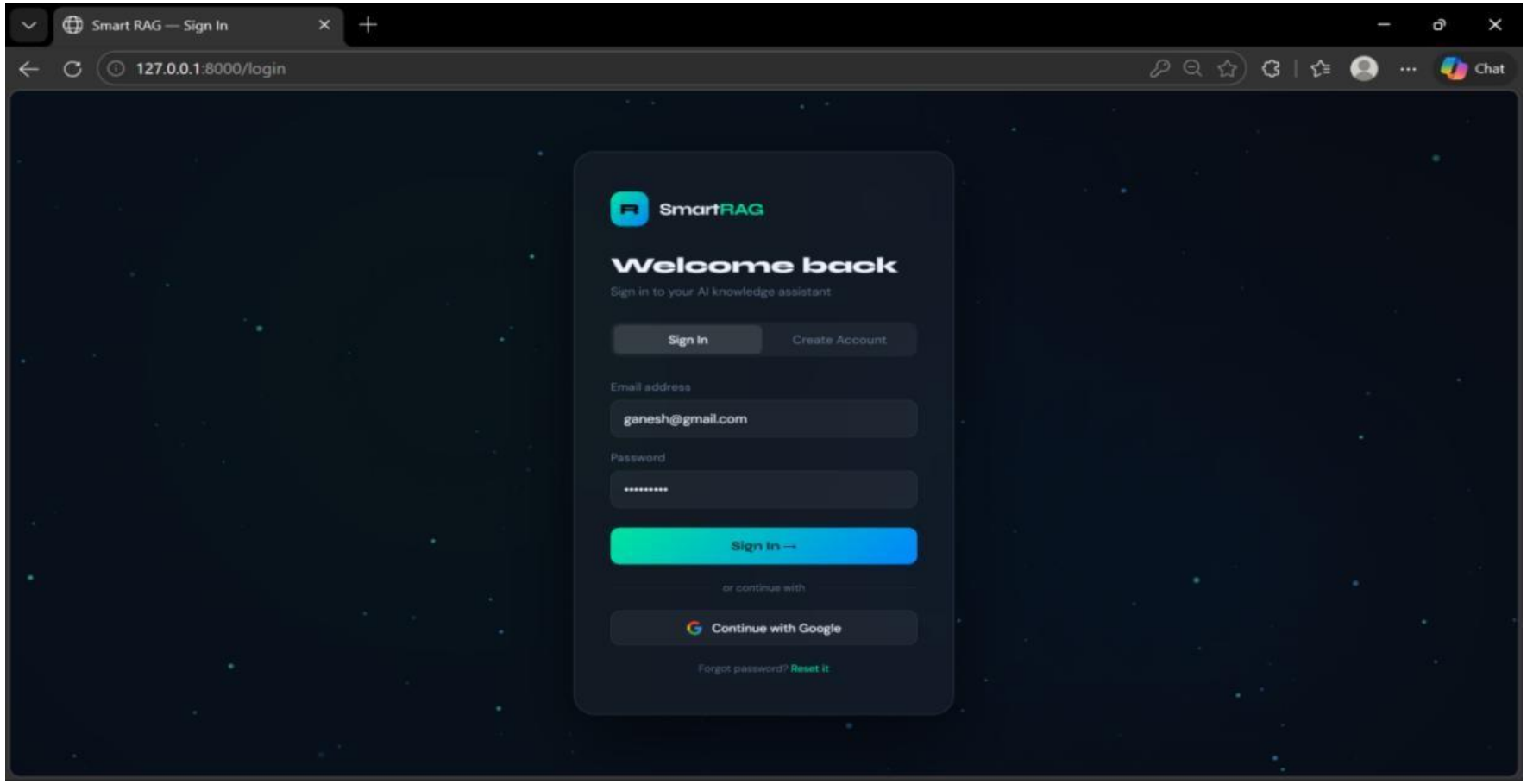
SYSTEM TESTING

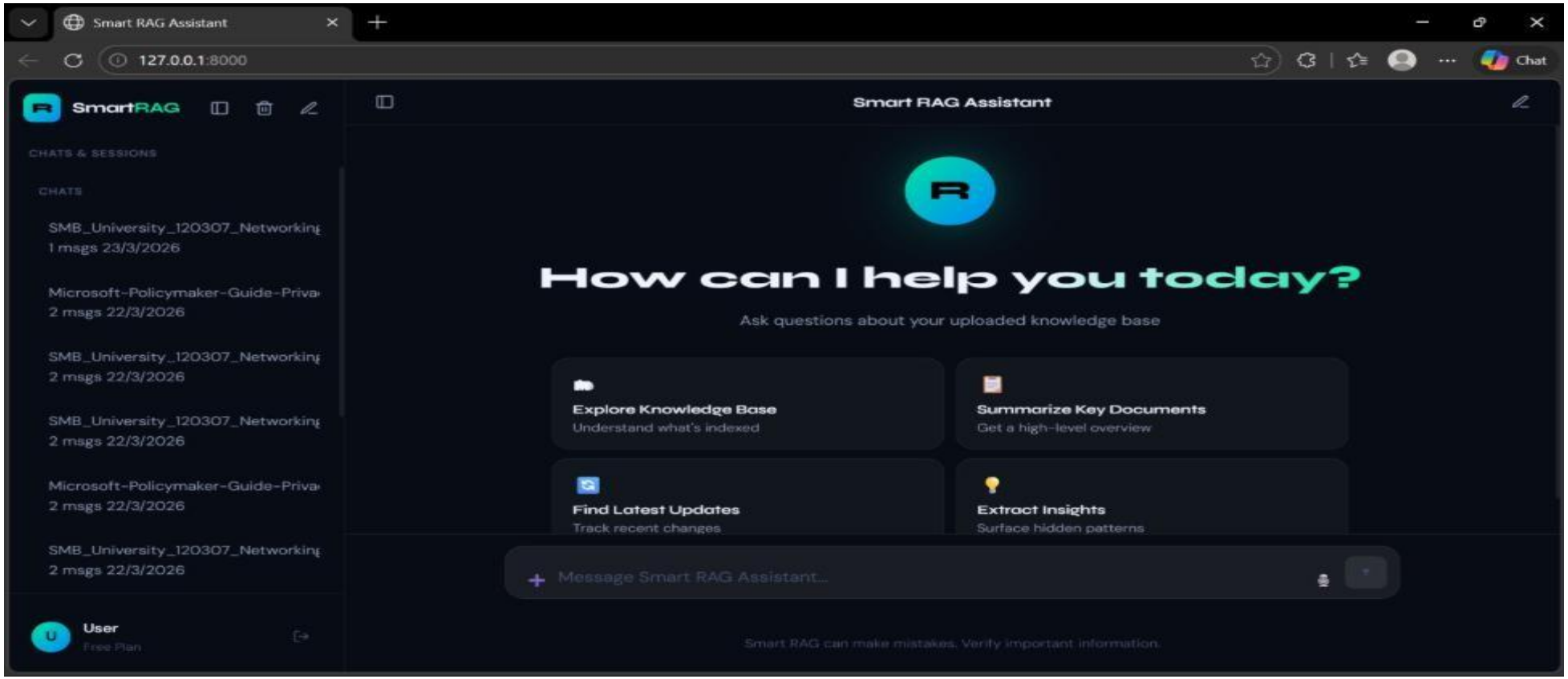
- Testing phase ensures reliability, accuracy, and performance of the system
- Validates functionality and identifies errors before deployment

Testing Types:

- **Unit Testing:** Tests individual modules
- **Integration Testing:** Ensures modules work together
- **Functional Testing:** Verifies system functions as expected
- **Performance Testing:** Checks speed and efficiency
- **Security Testing:** Ensures data safety and protection
- **Validation Testing:** Confirms system meets requirements

screenshot


Smart RAG — Sign In
127.0.0.1:8000/login
Chat
SmartRAG
Welcome back
Sign in to your AI knowledge assistant
Sign In
Create Account
Email address
ganesh@gmail.com
Password
Sign In →
or continue with
Continue with Google
Forgot password? Reset it


Smart RAG Assistant
127.0.0.1:8000
Chat
SmartRAG
CHATS & SESSIONS
CHATS
SMB_University_120307_Networking
1 msgs 23/3/2026
Microsoft-Policymaker-Guide-Priva
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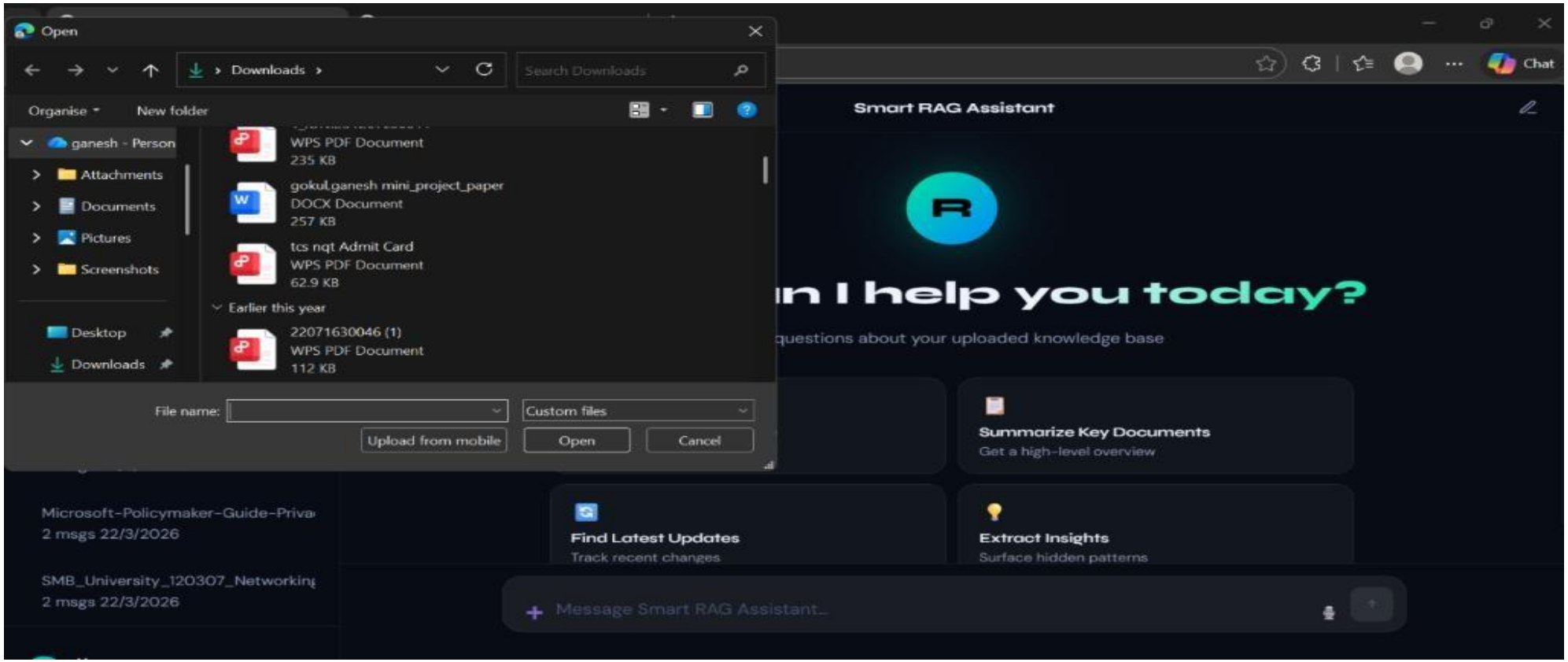
Microsoft-Policymaker-Guide-Priva
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
User
Free Plan
Smart RAG Assistant
How can I help you today?
Ask questions about your uploaded knowledge base
Explore Knowledge Base
Understand what's indexed
Summarize Key Documents
Get a high-level overview
Find Latest Updates
Track recent changes
Extract Insights
Surface hidden patterns
Message Smart RAG Assistant...
Smart RAG can make mistakes. Verify important information.


Open
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gokul,ganesh mini_project_paper
DOCX Document
257 KB
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112 KB
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Chat
Smart RAG Assistant
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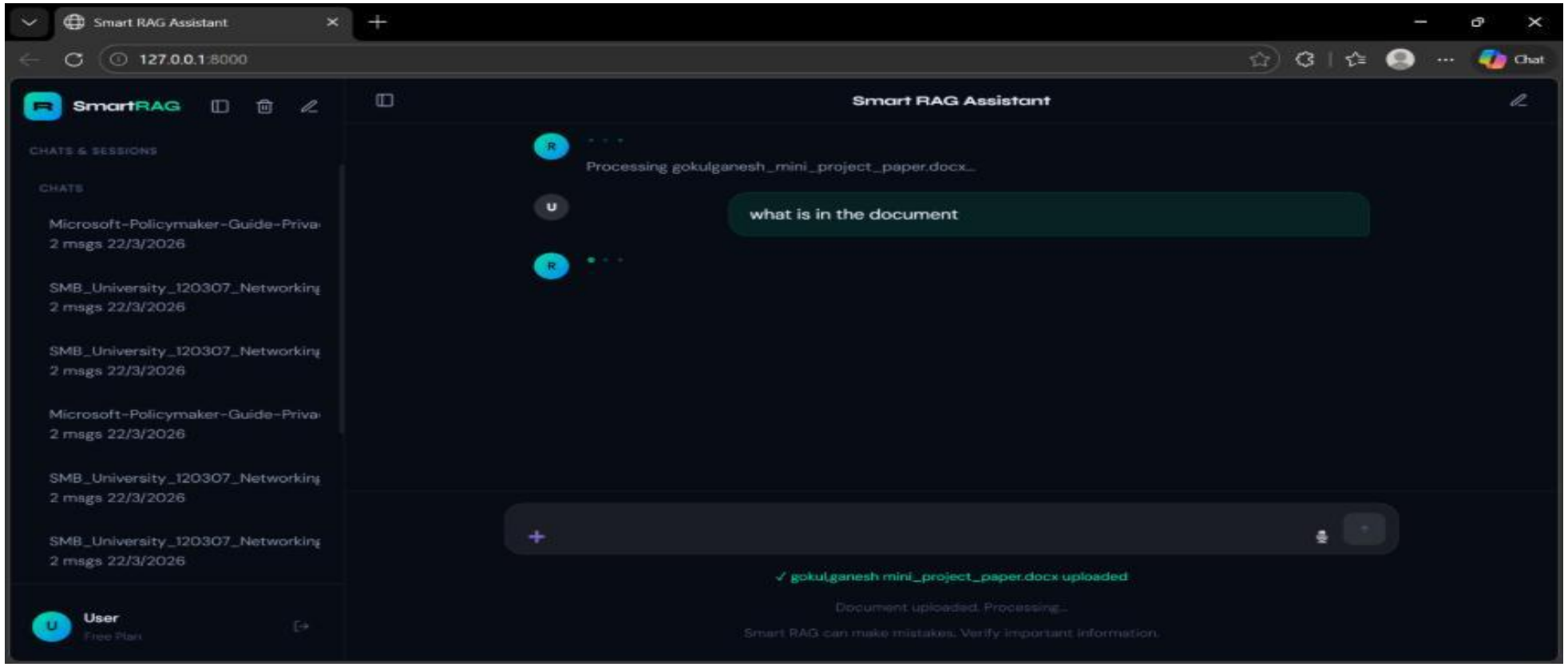
questions about your uploaded knowledge base
Summarize Key Documents
Get a high-level overview
Find Latest Updates
Track recent changes
Extract Insights
Surface hidden patterns
Microsoft-Policymaker-Guide-Priva
2 msgs 22/3/2026
SMB_University_120307_Networking
2 msgs 22/3/2026
Message Smart RAG Assistant...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SMB_University_120307_Networking
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User
Free Plan
Smart RAG Assistant
Processing gokulganesh_mini_project_paper.docx...
what is in the document
✓ gokulganesh mini_project_paper.docx uploaded
Document uploaded. Processing...
Smart RAG can make mistakes. Verify important information.

clustering algorithm that can automatically classify players into teams based on their jersey colors, reducing the need for manual labeling and increasing the efficiency of the analysis process.

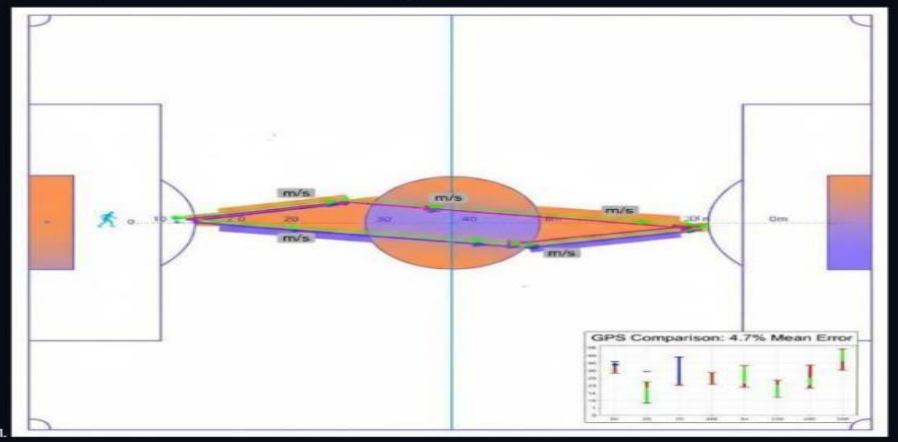
4. **Integration of multiple technologies**: The system combines YOLO object detection, homography transformation, and optical flow estimation to provide a comprehensive analysis of football matches, making it a robust and reliable solution.
5. **Potential for future enhancements**: The paper highlights several areas for future research and development, including the use of multi-camera fusion techniques, action recognition models, and real-time dashboard interfaces with tactical visualization tools, which could further enhance the system's capabilities and accuracy.

U

what are image in this paper

R

Based on the provided context, there are four images available from the document:



1. **Image Description:** A figure showing the YOLO football player detection result, with bounding boxes for players, referee, and ball.

CONCLUSION

- Developed a Smart Answering System using RAG for accurate and context-aware responses
- Combines retrieval and AI models to overcome limitations of traditional systems
- Provides fast, reliable, and meaningful answers in real time
- Achieves high accuracy, efficiency, and user-friendly interaction
- Suitable for applications like education, customer support, and knowledge systems
- Scalable system with future scope for multimodal and real-time enhancements

FUTURE ENHANCEMENTS

- **Cloud Integration:**

  Integrate with AWS, Azure, or Google Cloud for scalable data handling

- **Collaboration:**

  Use real-world network data from organizations to improve accuracy

- **Scalability:**

  Enhance system to handle large-scale networks (banking, healthcare, telecom)

- **Visualization:**

  Add advanced charts, heatmaps, and predictive analytics for better insights

THANK YOU!